

Skills Summary

The Zero That Holds a Place

Use this reference while completing your worksheet or when studying.

Skill 1 — Expanded form when a place is empty

Rule: Write a plus part only for places that hold something. An empty place (a zero digit) gets *no* part in expanded form — but its zero must still appear in the standard form.

Example: Write 308 in expanded form.

Step 1. Read each place: three hundreds, an empty tens place, eight ones.

Step 2. Write parts only for the places that hold something: 300 and 8. $\Rightarrow$ $308 = 300 + 8$

Try it: Write 902 in expanded form.

check: $902 = 900 + 2$

(!) Watch out: Expanded form of 308 is *not* $30 + 8$. The zero keeps the 3 in the hundreds place, so its part is 300.

Skill 2 — Standard form: keep the placeholder

Rule: To pack hundreds + ones back into one number, give *every* place a digit. A place with no part gets a **0**. Three-digit numbers always show three digits.

Example: Write $700 + 6$ in standard form.

Step 1. Match parts to places: 700 fills the hundreds, 6 fills the ones, and nothing fills the tens.

Step 2. Hold the empty tens place with a zero: seven, zero, six. $\Rightarrow$ **706**

Try it: Write $400 + 8$ in standard form.

check: 408

Skill 3 — Finding and fixing a placeholder error

Checklist for spotting the error:

- Add up the wrong work — what number does it *really* make?
- Compare it with the number it was supposed to make.
- Find the digit that changed places, then rewrite it correctly.

Example: Jade wrote $520 = 500 + 2$. Fix her expanded form.

Step 1. Test her work first: $500 + 2$ makes only 502, not 520, so something moved.

Step 2. In 520 the 2 sits in the tens place, so its part is 20, and the zero shows no ones.
 $\Rightarrow$ $520 = 500 + 20$

Try it: Rex wrote $103 = 100 + 30$. Write the correct expanded form of 103.

Check: $100 + 3$